RETAIL STORE FOOTFALL & PEAK HOURS ANALYSIS

Python | Pandas | Matplotlib | Seaborn | ReportLab

This report analyzes retail store customer footfall data to identify peak and off-peak hours, compare weekday vs weekend traffic, and provide data-driven staffing recommendations for store management.

Dataset Overview

Raw Records Collected: 22,430
Records After Cleaning: 22,298
Stores Covered: 3
Metrics Tracked: Entry Time, Exit Time, Dwell Time, Store ID
Analysis Granularity: Hourly, Daily, Weekly, Store-level

Objectives

1. Identify the busiest and quietest hours of store operation
2. Compare weekday vs weekend customer traffic patterns
3. Analyze customer dwell-time behavior across the store
4. Compare footfall performance across multiple store locations
5. Generate data-driven staffing recommendations by hour

1. Data Cleaning Summary

Before analysis, the raw dataset was validated and cleaned using the following checks:

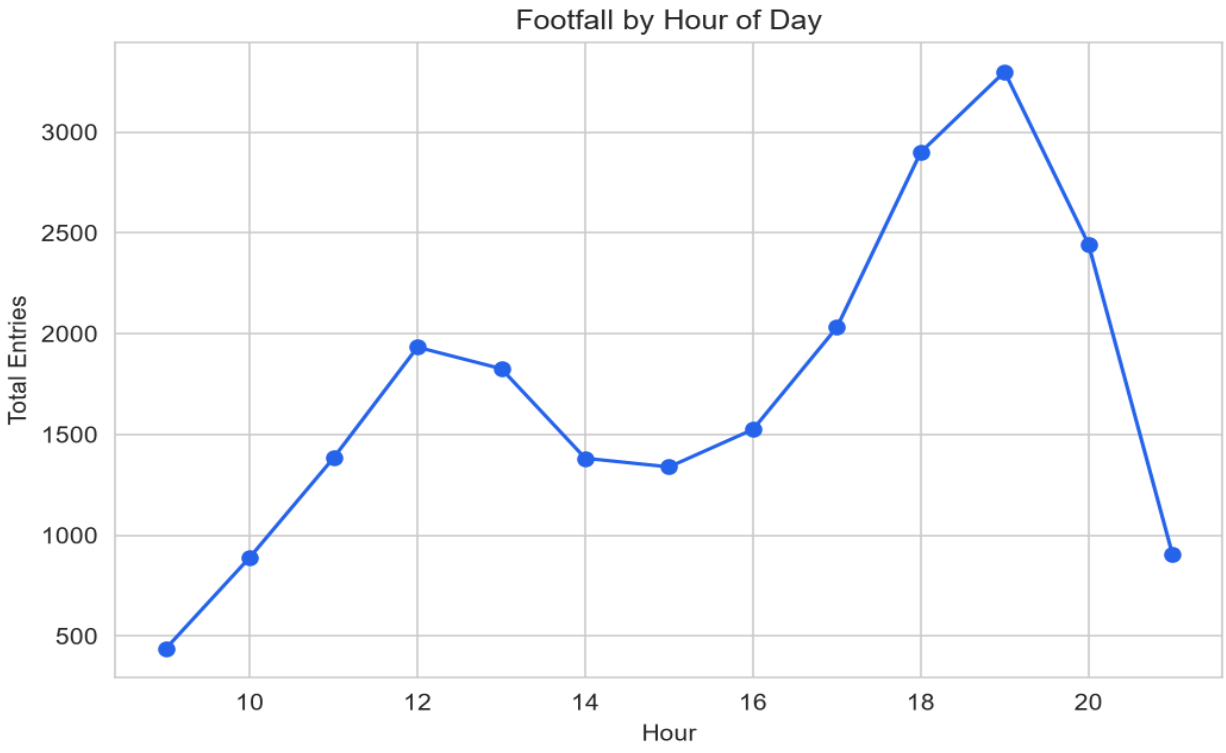
Check Performed	Rows Affected
Duplicate Visit_ID records removed	15
Missing Entry_Count values filled	237
Invalid dwell time / entry count removed	117
Invalid time-order (Exit before Entry) removed	0
Final clean record count	22,298

2. Executive Summary

Total Customer Entries: 22,298
Average Dwell Time: 24.79 minutes
Median Dwell Time: 25.0 minutes
Busiest Hour: 19:00
Quietest Hour: 9:00
Busiest Day: Saturday

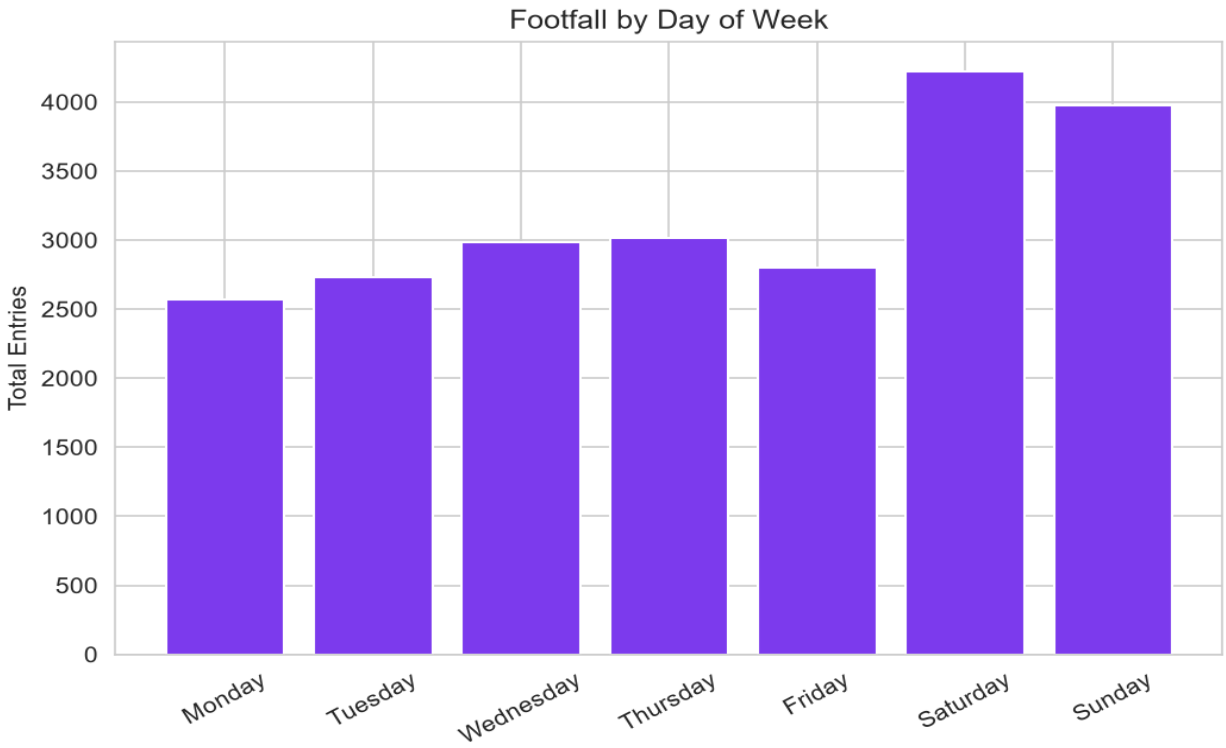
3. Hourly Footfall & Staffing Recommendation

Hour	Total Entries	Traffic Band
9	438	Low
10	890	Low
11	1,384	Low
12	1,933	Normal
13	1,826	Normal
14	1,382	Low
15	1,339	Low
16	1,525	Normal
17	2,031	Normal
18	2,900	Peak
19	3,299	Peak
20	2,444	Peak
21	907	Low



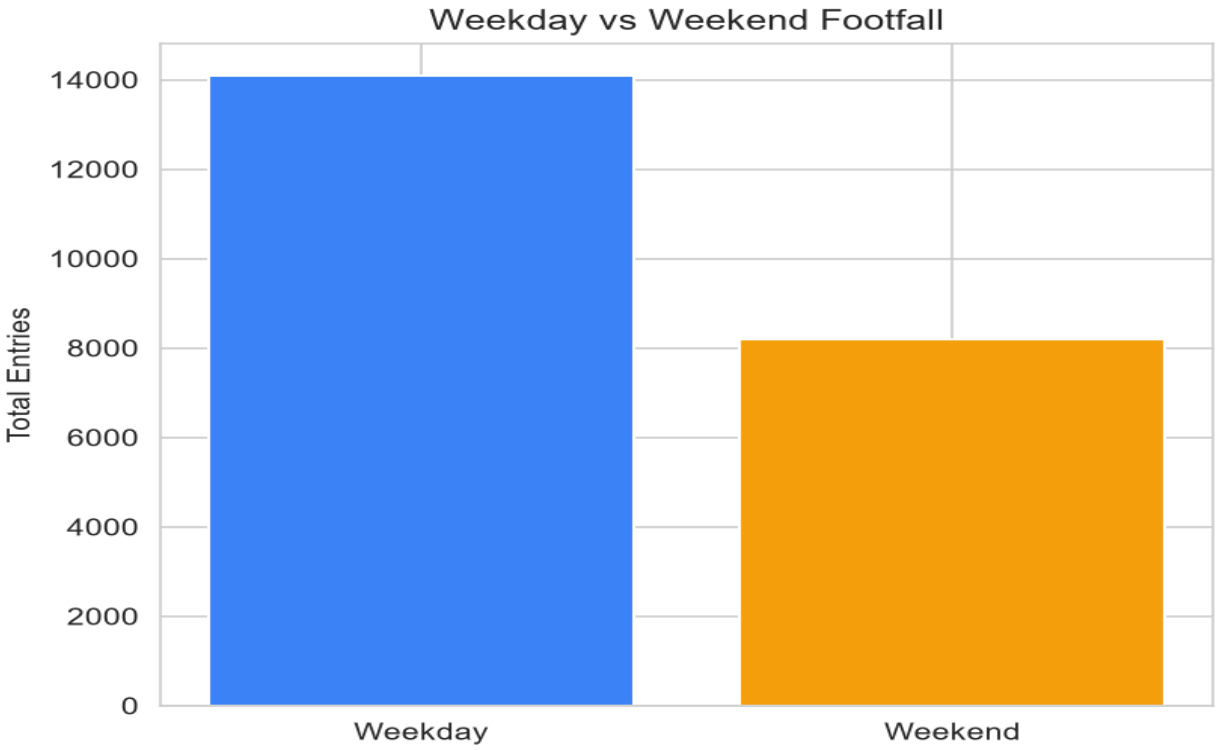
4. Footfall by Day of Week

Day	Total Entries
Monday	2,571
Tuesday	2,733
Wednesday	2,982
Thursday	3,012
Friday	2,798
Saturday	4,224
Sunday	3,978



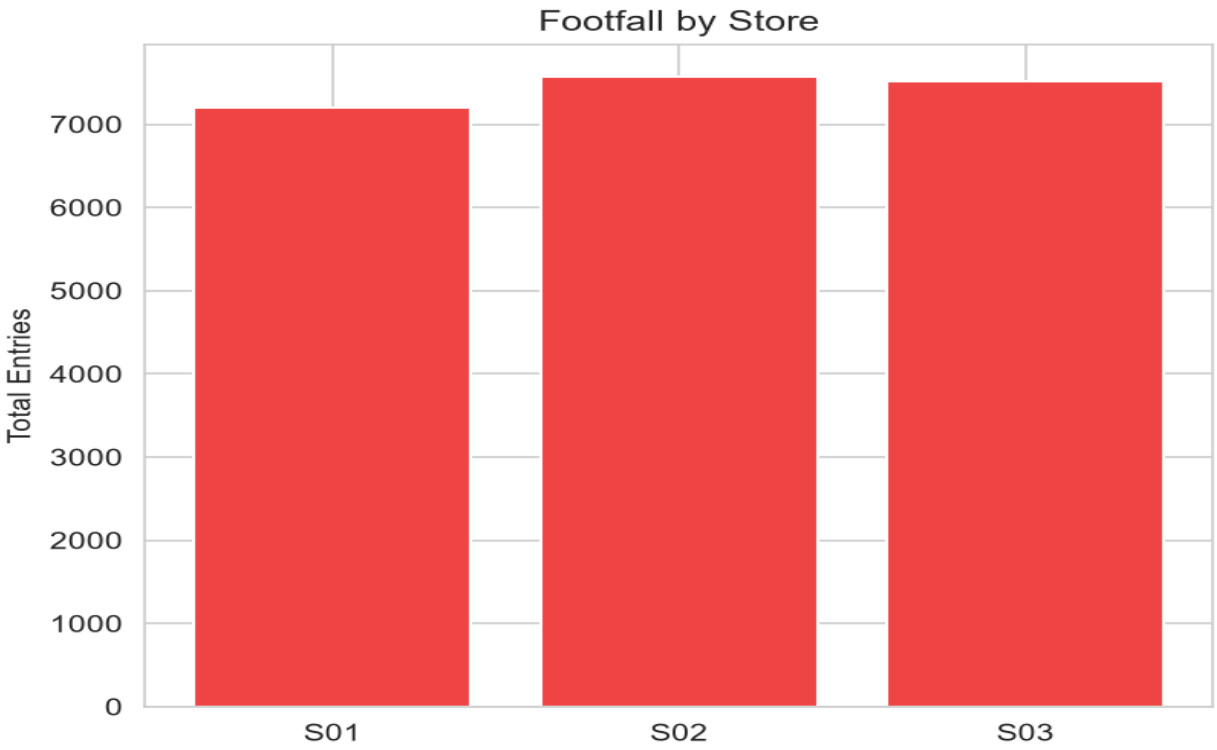
5. Weekday vs Weekend Comparison

Day Type	Total Entries
Weekday	14,096
Weekend	8,202



6. Store-wise Comparison

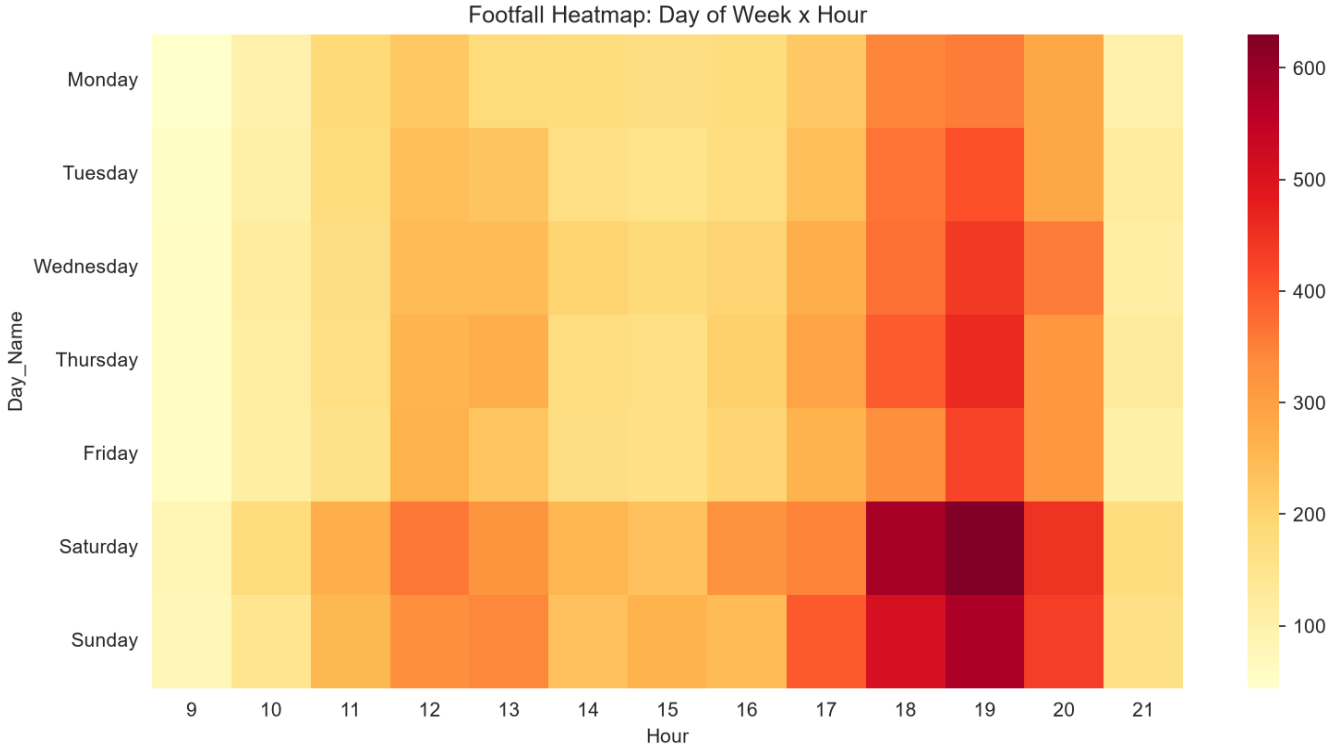
Store ID	Total Entries	Avg Dwell Time (min)
S01	7,206	24.67
S02	7,575	24.81
S03	7,517	24.87



7. Day x Hour Heatmap

This heatmap highlights exactly which day-hour combinations see the highest footfall. The table below lists the top 5 busiest day-hour combinations.

Day	Hour	Total Entries
Saturday	19:00	630
Saturday	18:00	582
Sunday	19:00	577
Sunday	18:00	510
Thursday	19:00	461

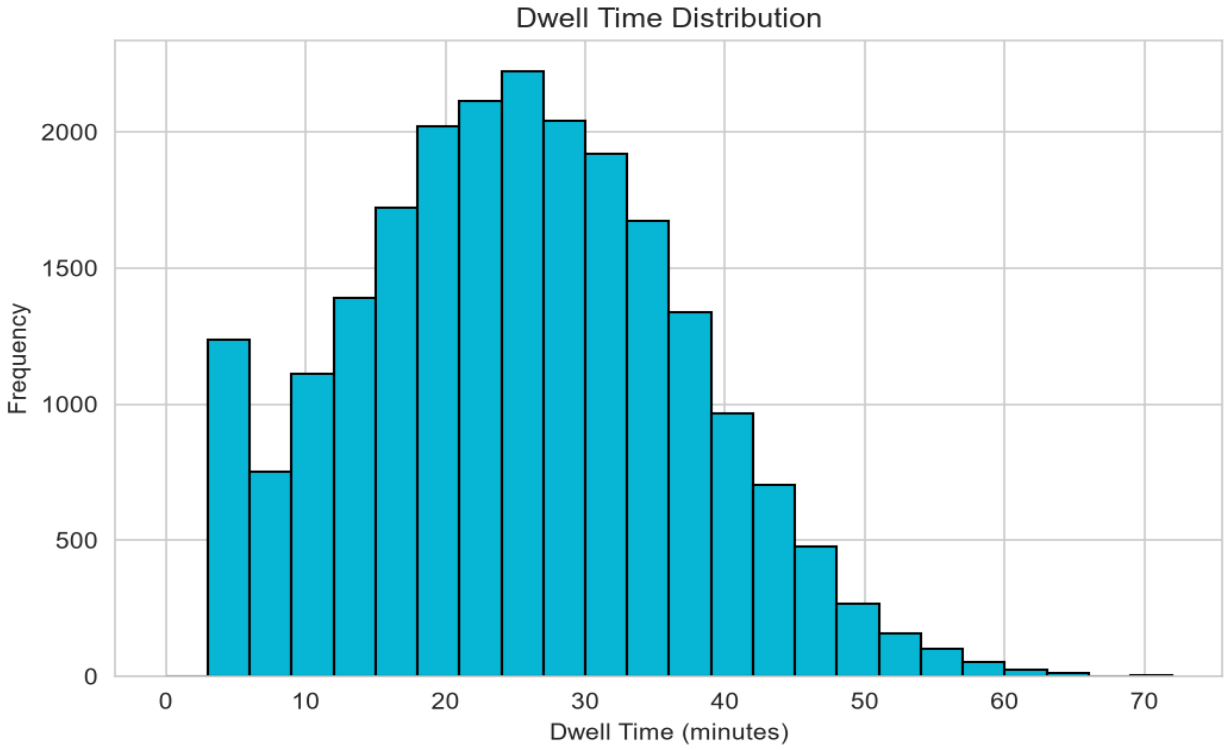


8. Dwell Time Analysis

Summary statistics by day type:

Day Type	Mean	Median	Min	Max	Std Dev
Weekday	24.72	24.0	3	71	11.67
Weekend	24.91	25.0	3	72	11.68

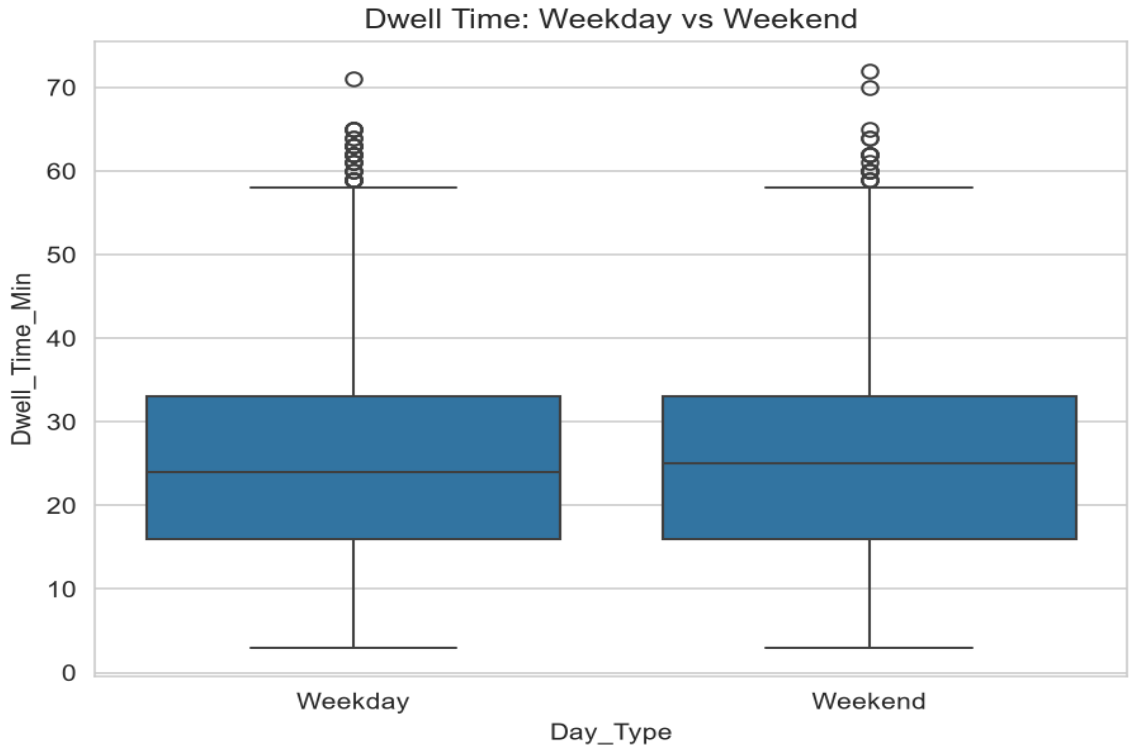
Distribution of how long customers stay in-store:



Dwell Time: Weekday vs Weekend Spread

The boxplot below compares the quartile spread of dwell time between weekdays and weekends. Key quartile values are summarized first:

Day Type	25th Percentile	Median (50th)	75th Percentile
Weekday	16.0	24.0	33.0
Weekend	16.0	25.0	33.0

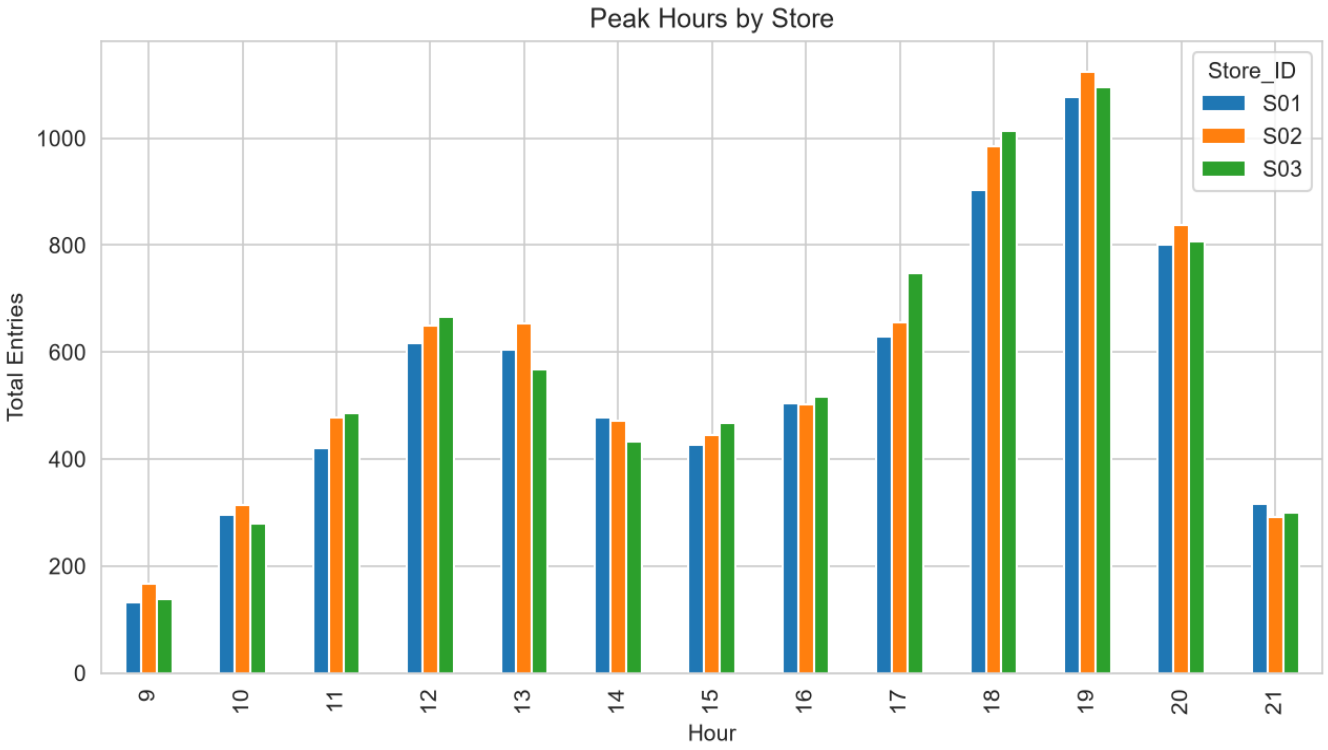


9. Peak Hours by Store

Each store's single busiest hour and its footfall:

Store ID	Peak Hour	Entries at Peak
S01	19:00	1,078
S02	19:00	1,125
S03	19:00	1,096

Each store's full hourly footfall pattern, compared side by side:

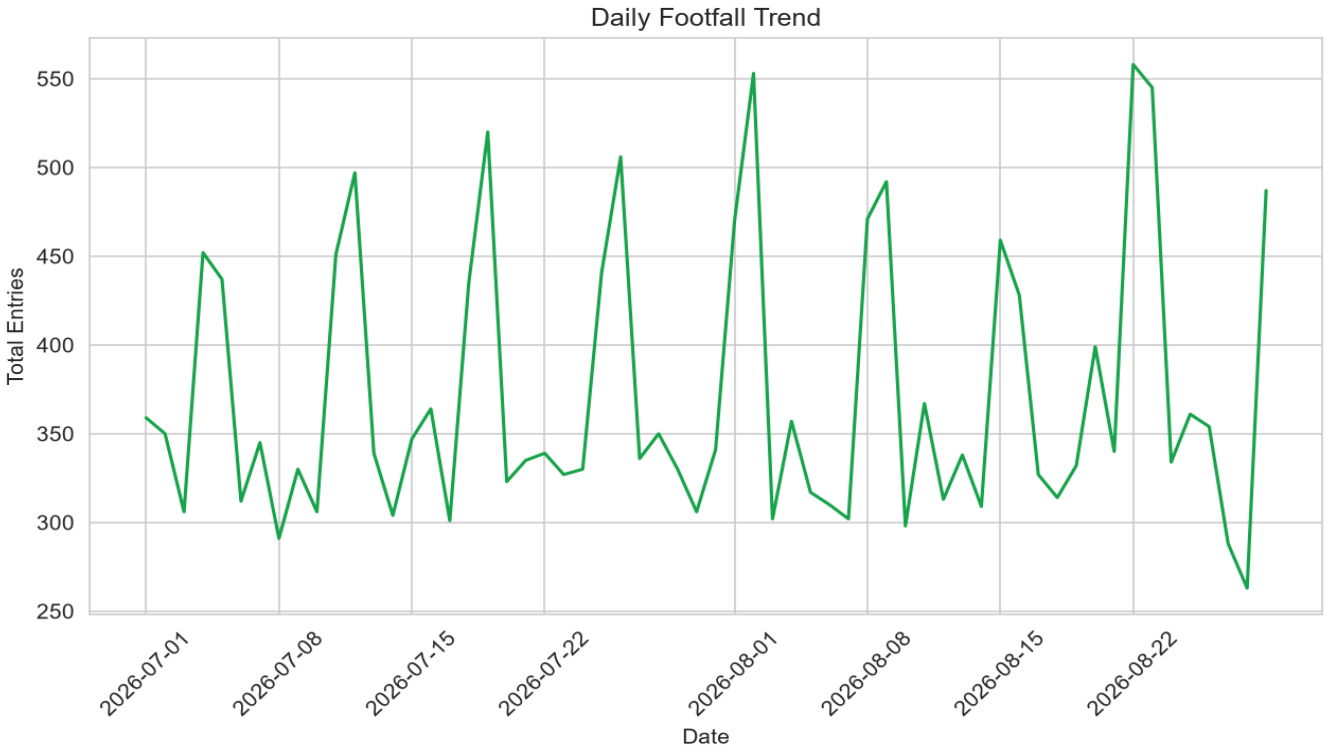


10. Daily Footfall Trend

Top 3 busiest and bottom 3 quietest days in the dataset:

Date	Total Entries	Type
2026-08-22	558	Busiest
2026-08-02	553	Busiest
2026-08-23	545	Busiest
2026-08-28	263	Quietest
2026-08-27	288	Quietest
2026-07-08	291	Quietest

Footfall trend across the full dataset period:



11. Key Insights & Staffing Recommendation

Busiest Hour: 19:00

Quietest Hour: 9:00

Busiest Day: Saturday

Weekend vs Weekday Gap: Weekend traffic is significantly higher per day than weekday traffic, indicating customer behavior shifts notably on Saturdays and Sundays.

Dwell Time Behavior: Average dwell time remains consistent at around 24.79 minutes regardless of day type, suggesting that while visitor volume changes, individual shopping duration does not.

Store Performance: All three stores show broadly similar footfall, with no single location significantly outperforming the others.

Recommendation: Increase staffing during hours flagged as "Peak" in the staffing table (Section 3), particularly between 6 PM and 8 PM on weekends, to reduce customer wait times and improve service quality. Conversely, staffing can be reduced during "Low" traffic hours (9 AM-11 AM, 9 PM) to optimize labor costs.

Report generated using Python, Pandas, Matplotlib, Seaborn, and ReportLab.